

MIAE 215 Assignment 1 Solutions – Winter 2023 Programming Tasks Premium

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Assignment #1 MIAE 215

Question #1

Write a program in C++ to print the code (ASCII code) of a given character.

Question #2

Given $3.4e38$ is the maximum float value (to within 1%), what is the output of the following program on a PC? Note: when cutting and pasting double quotes"" from Word you need to replace them in the cpp file since they don't copy properly.

```
int q;
float x,y,r = 5,v;
float r1 = 1.234567890123456, r2 = 1.234567890123455;
double w=1.0e19, d;
double u1 = 1.234567890123456, u2 = 1.234567890123455;

d = r1 - r2;
q = sizeof(float)*sizeof(int) - sizeof(double)*sizeof(char);
x = 4/3*3.14159*r*r;
y = 3.5*w*w;
y = y*0.5;
x = 7.7;
x += sin(x);
v = u1 - u2;

cout << d << "\n" << q << "\n" << x << "\n" << y << "\n" << v;
```

Question #3

a) What is the output of the following:

```
int i,j; double x=1.1, y=0.5, z=4.0;
if( (x/y) > 3 ) {
    z = abs(log10(z));
    z = sqrt(z*z);
    x = -x;
    y=pow(y,-x);
}

if( (x/y) <= 3 ) x = -x;
cout << x << "\t" << y << "\t" << z << "\n";
```

Mock exam

Summary

Quiz

Ask a question about this document

→



```
z = sqrt(1-x);
x = -x;
y=pow(y,-x);
}

if( (x/y) <= 3 ) x = -x;
cout << x << "\t" << y << "\t" << z << "\n";
```

```
for(i=1;i<=10;i++) cout << "\n" << i;
cout << "\ni = " << i << "\n";

j=sqrt(i);
cout << " " << j;
for(i=-1;i<=5;i=i+j) {
cout << "\n" << i;
if(i==3) i = 7;
}
```

b) Write a program that prints a table of t (1st column) and $\sin^2(t) + \cos^2(t)$ (2nd column) to the screen over a range from t = 0.0 to t = 3.14159 in 100 equal increments (i.e. make a table with 100 entries).

// e.g.	
// t	$\sin^2(t) + \cos^2(t)$
// 0.0	0.0
// dt	...
// 2*dt	...
// ...	...
// 3.14159	...

Question #4

What is the output of the following program? Note you need to include <cstdlib> to use exit(...). You also need to replace double quotes "" when cutting and pasting into a cpp file from Word.

```
int i, j;
double y=2.0000000000001, z;

z = 1.0 - cos(2*3.14159);

if(z == 0.0) {
    cout << endl << "z = 0.0";
}

if( abs(y - 2.0) < 1.0e-7 ) {
    cout << "\ny = 2.0";
}

if( sin(y) < 0.7 ) {
    cout << "\nsin(y) < 0.7";
} else {
    cout << "\nsin(y) >= 0.7";

    if( sin(y) > 0.95 ) {
        cout << "\nsin(y) > 0.95";
    } else {
        cout << "\n0.7 < sin(y) <= 0.95";
    }
}

}
```

```
i = 1;
for(j=-1;j<=15;j++) {
    i *= 10;
    if( i < 0 ) break;
}
cout << "\n" << j;
```

```
if( j > 5 ) {
    cout << "\nterminating program";
```



```
    } else {  
        cout << "\n0.7 < sin(y) <= 0.95";  
    }  
}  
}
```

```
i = 1;  
for(j=-1;j<=15;j++) {  
    i *= 10;  
    if( i < 0 ) break;  
}  
cout << "\n" << j;  
  
if( j > 5 ) {  
    cout << "\nterminating program";  
    exit(0);  
}  
  
cout << "\nprogram complete";
```

Question #5

Write a program that calculates $f = (\cos(t))^{0.5}$ in a for loop for $t=0, 0.1, 0.2, \dots, 1.0e7$ and stops the loop when $0.8 < f < 0.85$ (use just one if statement). Print out t and f when this occurs.

Question #6

Write a program that inputs a number x from the keyboard. It then divides x by two until $x < 1.0e-7$ or $\cos(1.0/x) < 0.2$. Print out the final values of x and $\cos(1.0/x)$ in scientific notation with a precision of 10 digits. Hint: use a for loop with a large final value for i . You could also make a loop involving x .

Question #7

Write a program that approximately determines the maximum value of the function

$$f = \sin(x^2x - 1.0) * (x - 3.0) + \cos(x)$$

over the range $0 \leq x \leq 10^3.14159$. This can be accomplished by evaluating the function from 0 to $10^3.14159$ at intervals of $dx = 0.01$ in a for loop (hint $x += dx$). For each loop iteration, if the function value f is greater than the variable m then set m to f . This process is repeated until the end of the loop (i.e. when $x > 10^3.14159$). The final value of m is equal to the maximum. Print this value to the screen. Set the initial value of m before entering the for loop to $f(0)$. Note this approximate maximization approach outlined above is essentially testing all the points at dx intervals in order to find out which one has the largest f .



**Question #8**

Write a program that approximately determines the first root of the function in the interval between 0 and 1.2:

$$x^2 * (x - 3.0) + \cos(x)$$

Hint: This can be accomplished by evaluating the function from 0 to 1.2 at intervals of $dx = 0.001$ in a for loop, and the approximation error of equal or less than 0.01 is accepted.